
Fixed-Spectrum Swaps Separate Attention Capacity from Assignment

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Abstract

1 Attention concentration cannot identify where a transformer routes information:
2 two attention rows may share the same weight spectrum while assigning their
3 largest weights to different tokens. This is a measurement validity problem. En-
4 tropy, the maximum weight, every norm, and the sorted spectrum are invariant to
5 token permutation, so none of them can vary with the quantity a routing claim is
6 about, and none can serve as evidence for it. We introduce the fixed-spectrum swap
7 (FSS), a causal intervention that preserves every permutation-invariant property of
8 an attention row while changing its assignment to task-labeled positions. Across
9 controlled retrieval, pretrained language models, and context-grounded question
10 answering, the intervention separates selective capacity from correct assignment
11 and reveals assignment-responsive, saturated, and selector-mismatch regimes. A
12 local value-aware utility term ranks finite margin effects in controlled retrieval with
13 0.989 Spearman correlation and 99.5% sign agreement, and utility-selected cir-
14 cuits yield positive held-out effects on natural question answering. Context-length
15 sweeps show where these diagnoses stop being sufficient: in some cases, behavior
16 degrades without corresponding routing changes. FSS gives a controlled test of
17 whether attention assignment causally affects a model’s prediction.

18 1 Introduction

19 Interpretability claims about transformers often rest on where attention concentrates. A sharper row,
20 a lower entropy, or a larger maximum weight is read as evidence that a model is attending to the right
21 evidence. We ask whether that inference is licensed. At fixed concentration, does assigning the same
22 attention spectrum to task-required evidence change the prediction?

23 The inference is not licensed, and the reason is structural. Entropy, the maximum weight, every norm,
24 and the full sorted spectrum are invariant to permutations of token positions. Two attention rows can
25 therefore share all of these statistics while one places its largest weight on the required evidence and
26 the other places it on a distractor. A measurement that takes the same value on both rows cannot be
27 evidence for either, so concentration is not an identifying measure of task-correct routing.

28 The question matters where routing accounts are used. When performance declines with length, a
29 transformer may learn a retrieval rule on short sequences and then lose it as distractors accumulate.
30 We see this pattern in copying, key-value recall, and algorithmic tasks [Kazemnejad et al., 2023,
31 Wang et al., 2024, Vasylenko et al., 2025]. As contexts grow, attention often becomes more diffuse,
32 entropy rises, and attention-output variance can collapse [Li et al., 2025]. These trends describe scale
33 and concentration, yet they do not tell us where the attention goes.

34 We test this with a fixed-spectrum swap (FSS). At the answer query, we exchange the weight on a
35 known source with a distractor weight and then run the unchanged model. Because the exchange
36 preserves the complete spectrum, a distractor-only exchange gives a matched control. This paired

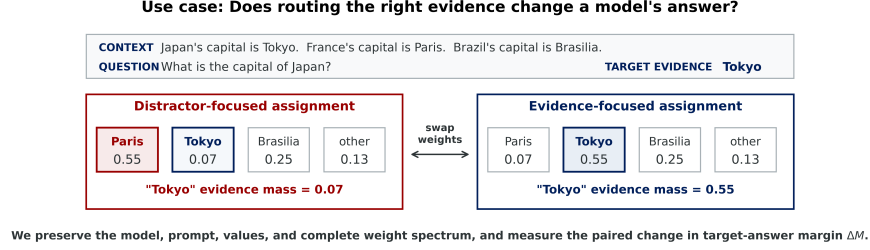


Figure 1: Question-answering use case. For the same context and question, a fixed-spectrum swap (FSS) moves the maximum attention weight from a distractor (“Paris”) to required evidence (“Tokyo”). The model, values, and weight spectrum remain unchanged; the paired target-answer margin tests whether assignment affects the answer.

design also lets us ask when moving weight to the source helps. The answer depends on the spectrum’s selective capacity, its assignment to task-labeled positions, and the downstream utility of the moved values.

Source-max raises per-token accuracy by 0.153, source-min lowers it by 0.043, and the distractor control changes it by 0.011. As the spectrum sharpens, the gap between the two assignments grows even though sharpening the original assignment remains near zero, and a local utility term orders the finite margin effect. Qwen and Pythia respond in the same direction, while Gemma and SmolLM2 expose saturation and selector-mismatch regimes.

Our contributions are:

- **Fixed-spectrum identification.** We separate selective capacity from task-conditioned assignment by preserving the complete attention spectrum and altering only its token placement.
- **Utility-conditioned mechanism.** We characterize when source-donor reassignment changes the final output and test the prediction against exact interventions.
- **Transfer and boundaries.** We test the intervention across controlled tasks, pretrained models, multi-evidence computation, and natural QA, and identify saturation and selector-mismatch regimes.

2 Related Work

Attention concentration and what it does not identify. Attention concentration has been connected to optimization, token selection, extrapolation, and sparse long-context computation [Tarzanagh et al., 2023, Wang et al., 2024, Nakanishi, 2025, Vasylenko et al., 2025]. Other accounts emphasize attention-output scale or normalization [Li et al., 2025, Zsámboki et al., 2025]. These statistics often move together as context length changes. They are used as evidence in routing accounts, but their status as identifying measures of task-correct routing is generally untested. Our intervention tests that identification question while the entire spectrum is held fixed.

Attention intervention and non-identifiability. Attention weights have long been studied as explanations, perturbed to test prediction sensitivity, and shown to be non-identifiable without their value vectors [Jain and Wallace, 2019, Wiegrefe and Pinter, 2019, Brunner et al., 2020, Kobayashi et al., 2020, Hassid et al., 2022]. Causal circuit analysis patches internal activations and measures the resulting output change [Yamakoshi et al., 2023, Conmy et al., 2023]. Attribution patching uses a gradient approximation to screen many candidate components efficiently [Syed et al., 2023, Kramár et al., 2024]. Our utility term has a related first-order form. Here we evaluate it along an exact permutation of a model’s own attention row, alongside the finite swap itself.

Closest attention interventions. The nearest methods delete, block, renormalize, or rank parts of an attention pathway: Attention Knockout blocks selected edges and renormalizes the row [Geva et al., 2023], efficient attention projects a row onto its output-effective subspace [Naim and Asher, 2024],

and Contribution Weights rank tokens by attention mass combined with projected-value magnitude, testing importance through removal [Cunningham and Muca Cirone, 2026]. Each changes the weight multiset. Our reassignment leaves it intact and changes only its assignment to token positions, and Equation 5 connects that finite, task-labeled swap to the target margin. Sparse and retrieval-based architectures instead change how distant context is accessed [Vasylenko et al., 2025, Hu et al., 2025]; we study a trained row with a fixed spectrum and isolate the effect of assigning its available selectivity to task-required evidence.

3 Methods

3.1 Capacity and Assignment

At an answer query, let $a = (a_1, \dots, a_n)$ be an attention row over n token positions, where $a_j \geq 0$ and $\sum_j a_j = 1$. Let $S \subseteq \{1, \dots, n\}$ contain the $r = |S|$ task-required evidence positions, and let $a_{(1)} \geq \dots \geq a_{(n)}$ denote the same weights in descending order. Define

$$C_r^+ = \sum_{i=1}^r a_{(i)}, \quad C_r^- = \sum_{i=n-r+1}^n a_{(i)}, \quad (1)$$

$$K_r(a) = C_r^+ - C_r^-.$$

C_r^+ and C_r^- are the largest and smallest evidence masses available under permutations of the fixed spectrum, and $K_r(a)$ is the available routing range. An assignment is a permutation π of $\{1, \dots, n\}$ that places weight $a_{\pi(j)}$ at token position j . Its evidence mass and normalized assignment coordinate are

$$m_S(\pi) = \sum_{j \in S} a_{\pi(j)}, \quad (2)$$

$$\rho_S(\pi) = \frac{m_S(\pi) - C_r^-}{K_r(a)} \in [0, 1], \quad K_r(a) > 0.$$

Thus $m_S(\pi) = C_r^- + \rho_S(\pi)K_r(a)$. If $K_r(a) = 0$, every assignment has the same evidence mass.

Capacity specifies how much evidence mass the spectrum can support.
Assignment specifies where that mass is placed.
Utility specifies whether moving it changes the answer.

Proposition 1 (Fixed-spectrum capacity-assignment decomposition). *For every assignment π of a fixed weight multiset,*

$$C_r^- \leq m_S(\pi) \leq C_r^+, \quad (3)$$

and both bounds are attained. If $K_r(a) > 0$, Equation 2 defines a unique $\rho_S(\pi)$. No statistic invariant to token permutation can determine ρ_S .

Proof sketch. An r -position evidence set receives at most the r largest weights and at least the r smallest. Assigning those weights to S attains each endpoint. Normalizing evidence mass within this interval gives ρ_S . Any permutation-invariant statistic is constant across all assignments, including assignments at opposite endpoints. The supplementary material gives the full exchange argument and the corresponding affine margin range. The decomposition is algebraic for every r . Behavioral validation is positive for $r = 1$ and $r = 2$; the $r = 3$ interval includes zero and the $r = 4$ models miss competence, so efficacy at higher arity is unestablished.

3.2 Utility-Conditioned Output Effects

Capacity and assignment describe attention mass. Prediction also depends on values and downstream computation. Let H be the selected attention heads. In each $h \in H$, let s be the evidence position, d_h the donor position whose weight is exchanged with it, and $a_{h,j}$ the original attention weight on position j . Define the transferred mass $\delta_h = a_{hd_h} - a_{hs}$. Let $x_{h,j}$ be token j 's residual-stream contribution after head h 's value and output projections. If z is the unpatched residual state, the intervention gives $z' = z + \Delta z$, where

$$\Delta z = \sum_{h \in H} \delta_h (x_{hs} - x_{hd_h}). \quad (4)$$

Let $\ell_k(z)$ be the downstream logit for token k , with target y and competitor c . Define $M_c(z) = \ell_y(z) - \ell_c(z)$, the exact change $\Delta M_c = M_c(z + \Delta z) - M_c(z)$, and local utility $u_{hj}^{(c)} = \nabla_z M_c(z)^\top x_{hj}$.

Proposition 2 (Utility-conditioned swap effect). *If $\nabla_z M_c$ is L_c -Lipschitz on $\{z + t\Delta z : t \in [0, 1]\}$, then*

$$\Delta M_c = \sum_{h \in H} \delta_h (u_{hs}^{(c)} - u_{hd_h}^{(c)}) + R, \quad |R| \leq \frac{L_c}{2} \|\Delta z\|_2^2. \quad (5)$$

For an affine downstream margin, $R = 0$.

The Taylor remainder is standard. The new component is the task-labeled capacity-assignment coordinate coupled to the exact FSS vector in Equation 4.

The intervention transfers attention mass from a donor token to the source. It helps when the source contributes more positively to the target margin than the displaced donor, unless downstream curvature overwhelms that gain. Source-max can therefore succeed, saturate, or fail under the same attention-mass rule. The supplementary material derives Equation 5, extends it to evidence sets, and states a sufficient margin certificate.

4 Experimental Setup

Controlled tasks. Argmax retrieval presents key-value pairs and requests the value paired with the largest key. Flag retrieval presents values with exactly one marked position and requests that value. Both have one labeled source. Two-evidence modular sum marks two digits and requests their sum modulo ten, so both sources are required. Four-layer decoders are trained with NoPE and RoPE and four seeds per cell; evaluation reaches $50\times$ training length. An eight-layer, width-512 grid checks scale. The inverse temperature $\beta \in \{1, 2, 4\}$ changes attention capacity while each assignment comparison preserves the resulting sorted spectrum.

Pretrained and natural-language tasks. In-context key-value recall lists pairs followed by a query key; the paired value is the labeled source. We test Qwen2.5, Pythia, Gemma, and SmolLM2 across increasing pair counts. The selector audit compares circuits ranked on disjoint calibration data by source mass or by output-conditioned utility. Natural MCQA uses four-choice questions with one gold passage and unrelated passages. Questions must be correct with the full context and gold passage alone, but incorrect without context. This filtering defines a model-specific context-dependent subset and is not intended to estimate average performance over the full dataset. Its length ladder freezes the four-passage circuit before forming nested 8-, 16-, and 32-passage contexts with additional unrelated passages.

Interventions and estimation. The distractor control swaps the largest and smallest non-source weights, preserving source mass while permuting the same attention row. For a source position s , let $d^+ = \arg \max_j a_j$, $d^- = \arg \min_j a_j$, and $D = \{j : j \neq s\}$ over valid tokens. Source-max swaps (s, d^+) and source-min swaps (s, d^-) . In controlled and pretrained key-value recall, the distractor control uses the maximum and minimum within D . Natural QA uses a displacement-matched control. With $\delta = a_{d^+} - a_s$, it chooses

$$(p, q) = \arg \min_{\{j, k\} \subset D} |a_j - a_k| - \delta \quad (6)$$

and swaps (p, q) . Among all non-source pairs, this choice minimizes the absolute difference between the control’s L_1 displacement and the treatment’s L_1 displacement; the remaining difference is the minimum attainable mismatch for that row. It holds source mass fixed, and every control preserves the complete spectrum. For natural QA, displacement is matched first because it measures perturbation magnitude. The control does not match the donor token, positional distance, or value norm, which remain separate confounds. Calibration and evaluation examples are disjoint, circuits are frozen before paired evaluation, and hierarchical intervals resample calibration or model seeds above paired examples. We interpret a cell only after its frozen competence gate passes. Three confirmatory contrasts form one inferential family: controlled source-max versus distractor control, the capacity-by-assignment interaction, and natural utility-selected source-max versus matched control. We use two-sided exact sign-flip tests at the trained-model or calibration-seed cluster

level and Holm correction across the family. Other intervals are descriptive and unadjusted. These swaps are counterfactual probes and need not lie on the model’s endogenous query-key manifold. Treatment and control share this intervention class, so their contrast measures downstream sensitivity to assignment, rather than the frequency with which the model naturally creates a particular row. Layer controls (+1.107 for selected $K = 8$ versus -0.059 for a random layer) and coverage dose response show greater sensitivity under this probe. They do not show that normal query-key computation produces swapped rows or that selected heads are the exclusive retrieval mechanism. In untouched distractor-order variants, the conditional source-mass coefficient is +1.250 ($p = .03125$), while the closest-spectrum paired effect is +.026 ($[-.017, +.070]$, $p = .25$). This connects the intervention to an endogenous association without establishing endogenous causality.

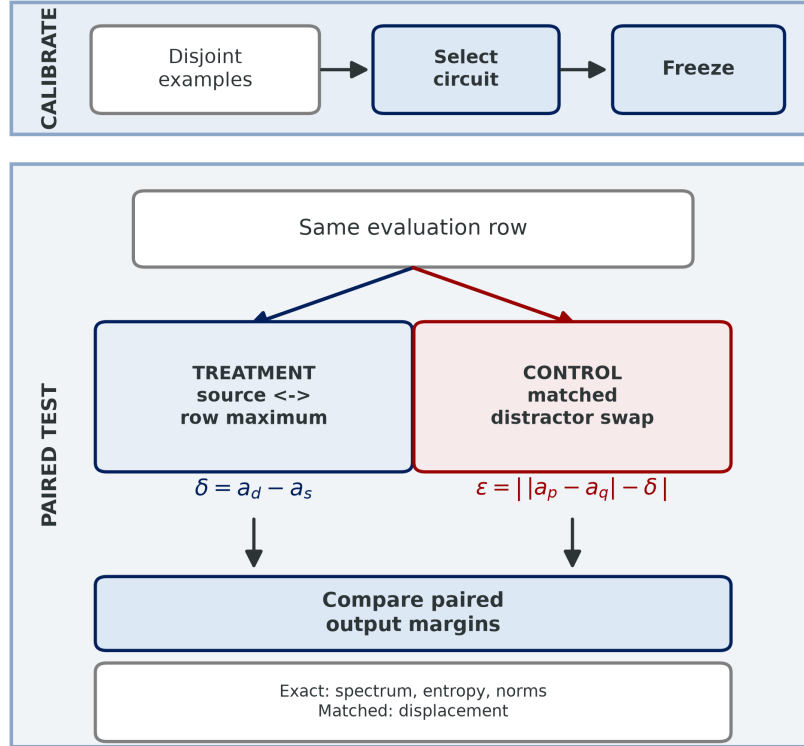


Figure 2: Paired causal protocol. A circuit selected on disjoint calibration data is frozen before treatment and control are applied to the same evaluation row. Both paths preserve the spectrum; the natural QA control additionally minimizes displacement mismatch in Equation 6.

Measured outcomes. We record task accuracy, target-versus-competitor margin, source mass, attention-output variance, entropy, norms, and the full sorted spectrum. Every fixed-spectrum condition preserves the selected attention rows to numerical precision. The supplementary material reports prompts, hyperparameters, per-length tables, competence failures, and preregistered decision rules.

5 Results and Discussion

5.1 Assignment Changes Accuracy at Fixed Spectrum

We begin with the simplest test: move the largest available weight to the source, move it away, or exchange two distractors. Across sixteen trained models, the mean changes are +0.153, -0.043 , and +0.011, respectively (Figure 3). Source-max beats the control in every model-length comparison.

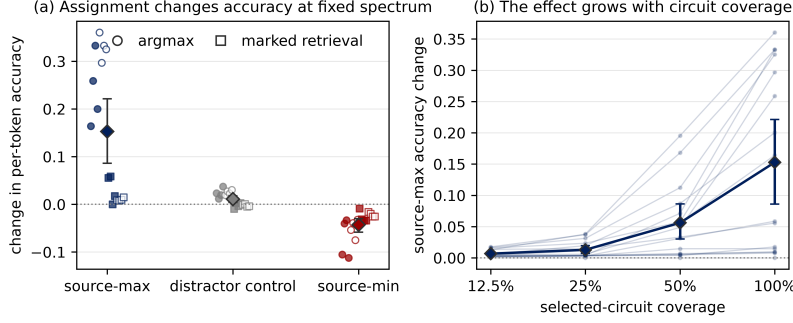


Figure 3: Assignment changes retrieval despite an identical attention spectrum. Source-max improves accuracy, source-min harms it, and the distractor-only control remains near zero (a). The effect increases with the fraction of the selected circuit receiving the intervention (b). Diamonds and intervals summarize trained models; faint paths show individual cells.

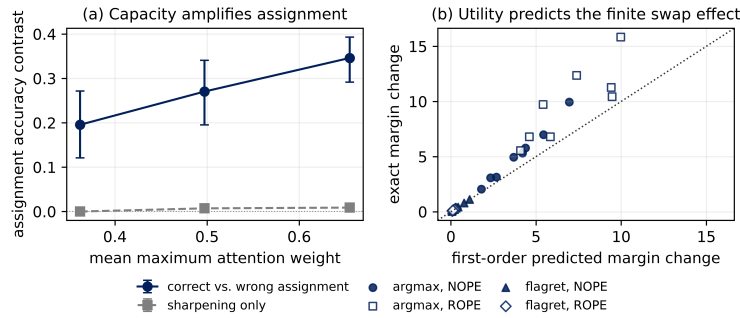


Figure 4: Capacity controls leverage, and utility predicts the finite intervention. Sharper spectra increase the correct-versus-wrong assignment contrast while sharpening alone remains near zero (a). The first-order utility term predicts the exact source-max margin effect across tasks and positional encodings (b). Error bars summarize trained models.

176 Their primary cluster-level contrast is $+0.142$ (95% CI $[+0.080, +0.206]$; exact $p = 3.05 \times 10^{-5}$;
177 Holm-adjusted $p = 9.16 \times 10^{-5}$), while complete-spectrum invariant error stays below 10^{-6} .

178 The controlled audit contains 8,192 head-example pairs. Individual swaps are vacuous in 18.5% of
179 them, but only 86 of 1,024 complete eight-head circuits (8.4%) are entirely vacuous. Mean exact
180 margin change is $+3.891$ over all circuits and $+4.248$ (95% CI $[+3.863, +4.640]$) over the 938 active
181 circuits. The median per-head transferred mass is .025 (90th percentile .930), and effect size rises
182 across transfer quartiles. The supplementary displacement audit reports the full distribution.

183 The effect grows with circuit coverage (Figure 3b). Patching one, two, four, and eight selected
184 heads gives mean source-max gains of 0.007, 0.013, 0.056, and 0.153 and is monotone in 31 of 32
185 model-length comparisons, consistent with a distributed effect under the probe.

186 Source attention tracks accuracy more closely than attention-output variance. Normalizing that
187 variance leaves accuracy unchanged, whereas forcing mass onto the known source restores controlled
188 retrieval. Sharpening the original row has little effect.

189 The ordering survives the eight-layer replication, both positional encodings, four training seeds, and
190 two controlled tasks. Random-layer and adjacent-layer interventions remain near zero when the
191 selected circuit is held fixed. Full model-level intervals and the variance and patching results appear
192 in the supplementary material.

193 5.2 Capacity Amplifies Assignment

194 We next vary spectrum and assignment independently. Temperature scaling changes the maximum
195 available weight, and at every temperature we evaluate identity, source-max, source-min, and dis-

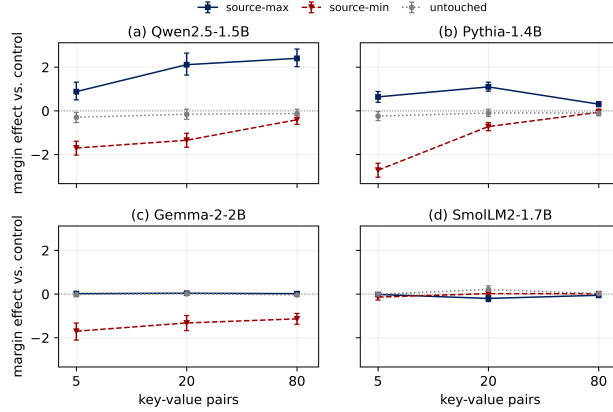


Figure 5: Matched pretrained-model comparison (seed 0; $n = 128$ paired examples per model-length cell). All panels use the 5-, 20-, and 80-pair ladder and report source-max, source-min, and untouched margin relative to the same distractor control. Qwen and Pythia respond positively, Gemma is saturated, and the source-mass-selected SmolLM2 circuit does not benefit. Error bars show paired 95% intervals.

tractor control on the same examples. The source-max minus source-min contrast grows from 0.196 to 0.346, while sharpening the identity assignment changes accuracy by at most 0.009 (Figure 4a). Capacity sets the possible leverage, while assignment determines where that leverage lands. The primary interaction is $+0.150$ (95% CI $[+0.094, +0.210]$; exact $p = 6.10 \times 10^{-5}$; Holm-adjusted $p = 1.22 \times 10^{-4}$).

For held-out examples we differentiate the target margin along the exact source-max path. The first-order term in Proposition 2 predicts the finite margin change with 0.917 Pearson correlation, 0.989 Spearman correlation, and 99.5% sign agreement (Figure 4b). The remaining error measures downstream curvature along the swap.

Modular sum requires the model to route and combine two marked values. At $5 \times$ training length, assigning the top two weights to evidence raises exact match by 0.066, assigning the bottom two lowers it by 0.291, and evidence-max exceeds a control that preserves both the spectrum and total evidence mass by 0.043 (95% CI $[0.017, 0.072]$). The result therefore extends beyond copying one token. A registered three-evidence estimate includes zero, and four-evidence models miss the frozen competence gate. Higher arity remains unresolved.

Evidence-max and evidence-min preserve the complete spectrum. The multi-evidence control also preserves total evidence mass, changing only which evidence token receives each weight. Its small effect separates aggregate source mass from useful evidence assignment.

5.3 The Same Intervention Produces Three Pretrained Regimes

The pretrained models show why the utility term matters. Qwen and Pythia gain margin when source-max replaces the distractor control. Gemma stays accurate as context grows and gains little from extra source mass, even though source-min reliably harms it. This is the saturation regime predicted by Proposition 2: removing evidence matters, while extra routing has little left to add. A quantized Qwen2.5-7B run preserves the positive margin direction.

SmolLM2 provides the contrasting case. Its preregistered source-mass selector fails, yet ranking heads on disjoint calibration examples by output-conditioned gradient or utility restores a positive effect. A ten-seed ablation ranks source gradient, utility gain, and utility gap above source mass and transfer mass. Rankings vary by architecture, but circuit selection must account for the output.

Interpolating from the natural row to the utility-selected swap gives a monotone margin path over five calibration seeds. On context-grounded natural QA, utility-selected routing improves margin over a displacement-matched control, while source-mass selection harms it. Across six disjoint calibration seeds, the utility-selected contrast is $+0.378$ (hierarchical 95% CI $[+0.253, +0.520]$; exact

Fixed-spectrum assignment		n Units	Unpatched	Δ source-max	Δ source-min	Δ distractor	max — distractor
Single evidence	$\beta = 1$	8,192 Acc.	.619	+.153	−.043	+.011	+.142
	$\beta = 4$	8,192 Acc.	.619	+.298	−.048	+.025	+.273
Two evidence	1×	2,048 EM	.952	+.019	−.559	+.012	+.007
	5×	2,048 EM	.484	+.066	−.291	+.023	+.043
Pretrained KV	Qwen2.5-1.5B	384 Margin	−3.048	+1.990	−.961	+.192	+1.799
	Pythia-1.4B	384 Margin	−3.805	+.826	−1.022	+.146	+.680
	Gemma-2-2B	384 Margin	+2.234	+.040	−1.373	+.013	+.027
	SmolLM2-1.7B	384 Margin	−2.782	−.168	−.112	−.077	−.091
Output-conditioned circuit selection		n Units	ΔM versus matched control		ΔM source mass	$\Pr(\Delta M > 0)$	utility — source
Task	Model		Utility gain	Utility gap			
Pretrained KV	Pythia-1.4B	384 Margin	+.519	+.359	+.006	.810	+.513
	SmolLM2-1.7B	384 Margin	+1.253	+.859	+.006	.969	+1.247
	Qwen2.5-1.5B	384 Margin	+1.493	+1.493	+.069	.956	+1.424

Table 1: Benchmark summary. The pretrained fixed-spectrum rows use the matched 5-, 20-, and 80-pair ladder from Figure 5, with 128 paired examples per model-length cell and $n = 384$ per model. Units make upper-block baselines comparable only within task. The lower block reports the three model families with completed matched selector runs; a Gemma selector-family run was not conducted. It has no unpatched baseline and evaluates selector quality using fixed-spectrum control effects, not per-row utility prediction; the supplementary held-out audit evaluates that prediction. $\Pr(\Delta M > 0)$ is the fraction of paired evaluation examples with a positive utility-gain source-max minus matched-control margin. Bold final columns remain identifiable in grayscale. Full cells and 95% intervals are in the supplementary material.

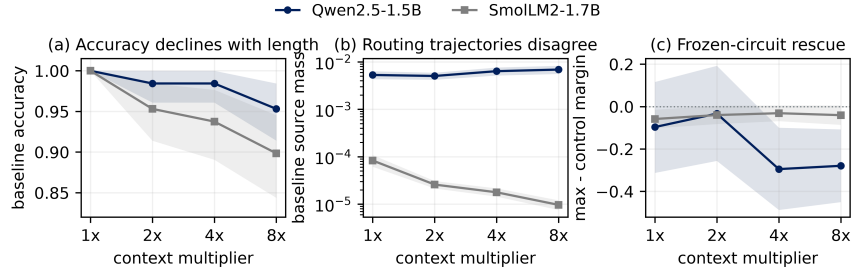


Figure 6: Matched natural-QA length ladder (seed 0; $n = 128$ paired examples per model-length cell). From 4 to 32 passages, accuracy falls (a), source routing moves in opposite directions (b), and frozen-circuit rescue stays flat or weakens (c). Bands are paired-example bootstrap 95% intervals.

228 and Holm-adjusted $p = .03125$). All competence gates pass. An unconstrained generation audit finds
229 no first-answer or repetition cost.

230 The selector results judge a selector by the frozen circuits it finds and their paired margin effects.
231 They do not test whether the per-row utility functional predicts a finite swap effect. That held-out
232 predictor audit is reported separately in the supplementary material.

233 Clean-to-corrupt activation patching rescues +.308 margin but only between 2.5% and 3.3% of
234 median corruption damage. It restores selected-head outputs after source-value corruption, so its
235 magnitude is not directly comparable to reassignment on an unchanged input.

236 5.4 Natural Context Marks the Scope Boundary

237 The fixed-context natural-QA result establishes causal routing in ordinary text. The length ladder is
238 a scope test: it asks whether the same fixed-context protocol remains informative as context grows.
239 We select the circuit at four passages, freeze every choice, and append unrelated passages to form
240 nested 4, 8, 16, and 32-passage contexts. The matched figure fixes seed 0 and resamples paired
241 questions; repeated passage observations from one circuit are not treated as independent models. The
242 supplementary material reports the additional Qwen seed separately.

Both Qwen and SmolLM2 lose margin and accuracy by 32 passages, yet their source-mass trajectories diverge. Qwen source mass rises as intervention rescue weakens. SmolLM2 source mass falls while rescue stays flat. The shared degradation cannot be attributed to one consistent source-mass trajectory. This is a scope limit of the diagnosis, not a failed prediction.

The protocol has a narrower interpretation. At a fixed context, a fixed-spectrum swap measures whether a chosen circuit can use reassigned evidence. As contexts grow, representation quality, distractor semantics, answer competition, and computation outside that circuit can also matter. The theorem concerns a specified swap at a fixed context; this experiment does not resolve the other long-context bottlenecks. Exact 32-minus-4 changes and hierarchical intervals appear in the supplementary material.

The arity checks provide another limit. The two-evidence task passes and supports the mass-preserving control; the three-evidence interval includes zero, and the four-evidence model misses competence. Higher arity remains unresolved. An eight-token generation audit also leaves first-answer accuracy and repetition unchanged, so simple format collapse is not driving the result.

5.5 Identification and Replication Audits

Replication checks vary the calibration sample, patch budget, and model family, and they support the same reading. Selected circuits are positive at every head budget, adjacent and random layers are smaller or near zero, the utility term ranks exact effects within every competent cell while calibrating them imperfectly, and an exact rerun reproduces the direction with an atypically large original magnitude. Appendix A reports these audits in full.

6 Limitations

FSS identifies the causal effect of a specified attention reassignment, not a complete account of model computation. The protocol requires identifiable evidence positions and a chosen intervention circuit; output-conditioned selection also requires disjoint calibration examples. The post-softmax swap is an off-manifold counterfactual. Preserving the full weight spectrum isolates assignment, but does not show that training would produce the same row or that the intervention is a practical inference-time policy.

Our empirical scope covers controlled retrieval and four pretrained families, including one quantized 7B replication. Natural QA is causal at fixed context, while divergent length-ladder trajectories show that the circuit does not explain every long-context failure. Higher arity remains competence-limited, the generation audit is short-horizon, and utility ranks swaps better than it calibrates large effects.

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A Identification and replication audits

Replication checks vary the calibration sample, patch budget, and model family. Across three frozen Qwen calibration seeds, selected circuits are positive at every head budget: $K = 2, 4, 8$ gives $+0.460$, $+1.054$, and $+1.107$ margin. Adjacent layers are smaller and inconsistent ($+0.147$ and -0.131), while random layers average near zero (-0.059). Full intervals and seed standard deviations are reported in the supplementary package.

A separate replication tests the utility relation. Circuits are chosen by source mass on disjoint calibration examples; utility is then computed on held-out evaluation rows. Across three seeds per family, every competent source-max cell is positive, and predicted effects rank the exact effects within each cell. Mean Spearman correlations are $.866$ for Qwen2.5-1.5B, $.544$ for Pythia-1.4B, and $.554$ for Gemma-2-2B, with sign agreements $.805$, $.807$, and $.714$. The approximation is imperfect at large effects, so we use it as a local ranking signal rather than a calibrated finite-effect estimator. For the 32 controlled cell means in Figure 4b, the actual-on-predicted fit is $1.387x - 0.102$ (Pearson $.978$), which deviates from identity calibration.

The exact Qwen rerun keeps precision, lengths, contrast, and sample counts fixed. All five new seeds are positive (mean $+0.042$); the combined six-seed test gives $p = .03125$, while the original $+1.882$ magnitude is atypical.

Output-conditioned selectors lead in each family, although their internal ranking changes by architecture. Full selector results, competence-gate failures, and the anonymized code package appear in the supplementary material.

B Theory details

This appendix gives the full statements and proofs behind Propositions 1 and 2, and a multi-evidence corollary. The complete per-cell result tables and secondary figures are in the separate supplementary material.

Established softmax grouping identity. For evidence positions S , distractors D , scores s_j , and temperature $T > 0$, let $Z_G = \sum_{j \in G} e^{s_j/T}$ and $F_G = -T \log Z_G$. Grouping the softmax denominator gives

$$m_S := \sum_{j \in S} a_j = \frac{Z_S}{Z_S + Z_D} = \sigma \left(\frac{F_D - F_S}{T} \right). \quad (7)$$

With the group log-mean-exp score $\bar{s}_G = T \log(|G|^{-1} \sum_{j \in G} e^{s_j/T})$,

$$m_S \geq \alpha \iff \bar{s}_S - \bar{s}_D \geq T \left(\log \frac{|D|}{|S|} + \text{logit}(\alpha) \right). \quad (8)$$

These Gibbs partition-function identities [Ramsauer et al., 2021] make the dependence on distractor scores explicit: distractor count alone is insufficient, since arbitrarily many negligible-score distractors need not reduce evidence mass.

Capacity-assignment decomposition. Let $a_{(1)} \geq \dots \geq a_{(n)}$, let S contain r evidence positions, and define $C_r^+ = \sum_{i=1}^r a_{(i)}$, $C_r^- = \sum_{i=n-r+1}^n a_{(i)}$, and $K_r(a) = C_r^+ - C_r^-$.

Theorem 1 (Capacity-assignment decomposition). *For every permutation π of a fixed attention-weight multiset, $C_r^- \leq m_S(\pi) \leq C_r^+$, both endpoints are attainable, and for $K_r(a) > 0$ the unique coordinate $\rho_S(\pi) = (m_S(\pi) - C_r^-)/K_r(a) \in [0, 1]$ gives $m_S(\pi) = C_r^- + \rho_S(\pi)K_r(a)$. For affine margin $M(\pi) = b + \sum_j a_{\pi(j)}u_j$ with sorted utilities $u_{(1)} \geq \dots \geq u_{(n)}$,*

$$\max_{\pi} M(\pi) = b + \sum_i a_{(i)}u_{(i)}, \quad \min_{\pi} M(\pi) = b + \sum_i a_{(i)}u_{(n+1-i)}. \quad (9)$$

No statistic invariant to token permutation can determine evidence assignment or the output margin.

Proof. An r -position set receives at most the r largest weights and at least the r smallest; assigning those weights to S attains the endpoints. Normalizing the bounded mass gives ρ_S . For the output

range, suppose $u_p > u_q$ but $a_{\pi(p)} < a_{\pi(q)}$. Swapping those weights changes the margin by $(a_{\pi(q)} - a_{\pi(p)})(u_p - u_q) > 0$. Repeated exchanges align weights and utilities for the maximum and reverse them for the minimum, the classical rearrangement inequality [Hardy et al., 1952]. The extremal assignments preserve the complete weight multiset and therefore every permutation-invariant statistic. $\square$

Nonlinear downstream network. Fix an input and selected heads H at one layer. Let $x_{hj} = O_h v_{hj}$ and $z = \sum_{h \in H} \sum_j a_{hj} x_{hj}$. For target y and competitor c , let $M_c(z) = g_y(z) - g_c(z)$. A source–donor swap has $\delta_h = a_{hd_h} - a_{hs}$ and $\Delta z = \sum_{h \in H} \delta_h (x_{hs} - x_{hd_h})$. Define local utility $u_{hj}^{(c)} = \nabla M_c(z)^\top x_{hj}$.

Theorem 2 (Nonlinear routing-margin bound). *If ∇M_c is L_c -Lipschitz from z to $z + \Delta z$, then*

$$\left| \Delta M_c - \sum_{h \in H} \delta_h (u_{hs}^{(c)} - u_{hd_h}^{(c)}) \right| \leq \frac{L_c}{2} \|\Delta z\|_2^2. \quad (10)$$

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Proof. The swap changes head h by exactly $\delta_h (x_{hs} - x_{hd_h})$. The fundamental theorem of calculus gives $\Delta M_c = \int_0^1 \nabla M_c(z + t\Delta z)^\top \Delta z dt$. Subtracting $\nabla M_c(z)^\top \Delta z$ and applying Lipschitzness bounds the remainder by $\int_0^1 L_c t \|\Delta z\|_2^2 dt = L_c \|\Delta z\|_2^2 / 2$ [Nesterov, 2004]. Expanding the first-order term gives Equation 10. $\square$

Corollary 3 (Multi-evidence routing). *Let S contain r evidence positions and pair every evidence position missing a top- r weight with one top- r non-evidence donor. For disjoint swap pairs P_h and $\delta_{hsd} = a_{hd} - a_{hs} \geq 0$, $\Delta z = \sum_{h \in H} \sum_{(s,d) \in P_h} \delta_{hsd} (x_{hs} - x_{hd})$, and Equation 10 holds with the first-order sum extended over $(s,d) \in P_h$. Assigning top- r capacity to the evidence improves the margin whenever the summed evidence–donor utility gain exceeds the curvature remainder.*

The decomposition and corollary are algebraic for arbitrary r . Their behavioral premise is established for $r = 1$ and $r = 2$; the three-evidence interval includes zero and the four-evidence models miss the competence gate, so efficacy beyond $r = 2$ remains open. The condition is sufficient, not necessary: source-max can fail when evidence values have no greater downstream utility, patched heads cancel, or curvature dominates, so natural source mass alone is not a universal circuit-selection rule.

406 C Experimental protocol

Controlled tasks and training. Argmax retrieval requests the value paired with the largest key; flag retrieval requests the single marked value; both are order-invariant with one identifiable evidence position. Reversed decimal addition is order-dependent contrast, and two-evidence modular sum marks two digits and requests their sum modulo ten so copying either digit is insufficient. The primary model is a four-layer pre-norm decoder transformer with width 256, eight heads, and feed-forward width 1024, trained with AdamW, batch size 512, and warmup plus cosine decay on lengths one through five. Evaluation follows a geometric ladder to $50\times$ training length, and the scale replication uses eight layers and width 512. A cell is interpreted only after training-length exact match reaches 0.8. Order-invariant cells use four seeds; the order-dependent contrast uses two. The preregistration fixes the competence gate, break-point rule, and variance-collapse check before intervention outcomes.

Measured quantities and interventions. At the answer query we record evidence mass, attention entropy, every attention norm, participation ratio, attention-output variance, and target margins. Source-max swaps the source weight with the row maximum, source-min swaps it with the row minimum, and the distractor control swaps the largest and smallest non-source weights. All conditions preserve model parameters, prompts, transformed values, the complete sorted attention row, entropy, norms, maximum, and participation ratio; the control additionally preserves source mass.

424 **Confirmatory inference.** One family of three primary contrasts is fixed in advance: controlled
425 source-max minus distractor control, the capacity-by-assignment interaction, and natural utility-
426 selected source-max minus matched control. Exact two-sided sign-flip tests operate on independent
427 trained-model clusters for the controlled claims and calibration-seed clusters for natural QA, and
428 Holm adjustment controls familywise error across the three tests. Hierarchical intervals and all other
429 reported cells are descriptive and unadjusted unless stated otherwise.

430 **D Scope, boundaries, and reproducibility**

431 **Scope.** The equals-sign second-prompt format fails its competence gate in Qwen, and a blinded
432 arrow-format pilot fails full-run competence despite directional margin signs; both are excluded
433 from causal replications. Pythia and Qwen are floor-limited at their longest contexts, so margin
434 is the interpretable tail outcome; Qwen-7B is quantized, and Gemma supplies a high-competence
435 saturation boundary. The fixed-spectrum theorem still applies to the natural-QA length result because
436 it conditions effects on transferred mass, value utility, and curvature, but the length ladder does not
437 support the stronger claim that natural length degradation must involve source-mass loss.

438 **Reproducibility.** The controlled tasks are generated from the included task generator and require
439 no external dataset. Natural MCQA samples are constructed from the public SQuAD training split
440 [Rajpurkar et al., 2016] with the sampling seed and eligibility rule included. Pretrained probes use
441 publicly released Qwen, Pythia, Gemma, and SmolLM2 checkpoints, with exact identifiers, tokenizer
442 settings, and precision recorded in the result artifacts. Pretrained experiments used a Google Colab
443 runtime with one NVIDIA T4 GPU, PyTorch, and Hugging Face Transformers; the earlier controlled
444 artifacts retain configurations and seeds but not the exact package-version snapshot, so we mark
445 infrastructure reproducibility partial. Saved artifacts reproduce every reported statistic and figure
446 without loading a language model, and an anonymized code and artifact archive accompanies the
447 submission as supplementary material.